

Computer Networks and Data Communication

Module 5

CS 212

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Module wise CONTENTS:

- **Module 1 - Introduction:** An overview of networks, the Internet, services and protocols, RFCs and standards of Networking, Data communication & its components, types of data communication and communication channels. Key components of computer network; network devices; types of computer networks; network topology; reference models (Network Architectures) – OSI & TCP/IP.
- **Module 2 - Physical Layer:** Introduction and functions; types of transmission media – guided and unguided; integrated services digital network (ISDN); asynchronous transfer mode (ATM).
- **Module 3 - Data Link Layer:** Overview; sliding window protocols - Stop-&-Wait; Go-Back-N, Selective Repeat; Error detection and correction; Hamming Code, LRC, VRC, CRC, Checksum, Round Trip Time, Latency (Delay); Medium Access Sublayer: medium access control (MAC) Addresses; Switching techniques - circuit, message & packet; Random Access Protocols - Pure ALOHA, Slotted ALOHA, CSMA / CD, CSMA / CA; Controlled Access Protocols – FDMA, TDMA, CDMA; Channelization Protocols – Reservation, Polling, Token Passing; Ethernet (802.3 IEEE standard), Wireless Links, Wireless LAN - WiFi (802.11).
- **Module 4 - Network Layer:** Introduction; address resolution protocol (ARP); internet protocol (IP); internet control message protocol (ICMP); internet group management protocol (IGMP); IPv4 addressing, representation of IPv6; IPv4 vs IPv6; Conversion IPv4 to IPv6; Routing Protocols - routing information protocol (RIP), open shortest path first (OSPF); border gateway protocol (BGP); Enhanced Interior Gateway Routing Protocol (EIGRP).
- **Module 5 - Transport Layer:** Introduction; transport layer protocols - UDP & TCP, TCP connection establishment; TCP connection termination; TCP congestion control. **Application Layer:** Introduction – features & functions; application layer protocols; Web and HTTP, Email, SMTP, P2P network, its types & applications.
- **Module 6 – Network Optimization, Mathematical Models, Introduction Advance Computer Networks, MANET, WSN, PCN, SDN, IoT.**

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- **Module 5 - Transport Layer:** Introduction; transport layer protocols - UDP & TCP, TCP connection establishment; TCP connection termination; TCP congestion control. **Application Layer:** Introduction – features & functions; application layer protocols; Web and HTTP, Email, SMTP, P2P network, its types & applications.

Transport Layer:

Introduction:

- The transport layer is positioned between the network and session layers in the OSI paradigm.
- **The data packets must be taken and sent to the appropriate machine by the network layer. After that, the transport layer receives the packets, sorts them, and looks for faults.**
- Subsequently, it directs them to the session layer of the appropriate computer program. Now, the properly structured packets are used by the session layer to hold the data for the application.

Transport Layer Protocols:

- Transport Layer Protocol uses different protocol for the better communication between two ends uses of protocol that may differ from specifications. Below mention are some protocols used in Transport Layer:

1. Transmission Control Protocol(TCP)

- TCP is connection-oriented Protocol.
- TCP is reliable protocol.
- As TCP is connection-oriented protocol, so first the connection is established between two ends and then data is transferred and then the connection is terminated after all data being sent.

Transport Layer Protocols:

2. User Datagram Protocol (UDP)

- UDP is not reliable protocol
- The protocol UDP is connectionless.
- When speed and size are more important than security and dependability, this kind of protocol is employed.
- The data from the higher layer is supplemented with transport-level addresses, checksum error control, and length information by UDP, an end-to-end transport level protocol.
- A user datagram is the packet that the UDP protocol generates.

Transport Layer Protocols:

3. Stream Control Transmission Protocol (SCTP)

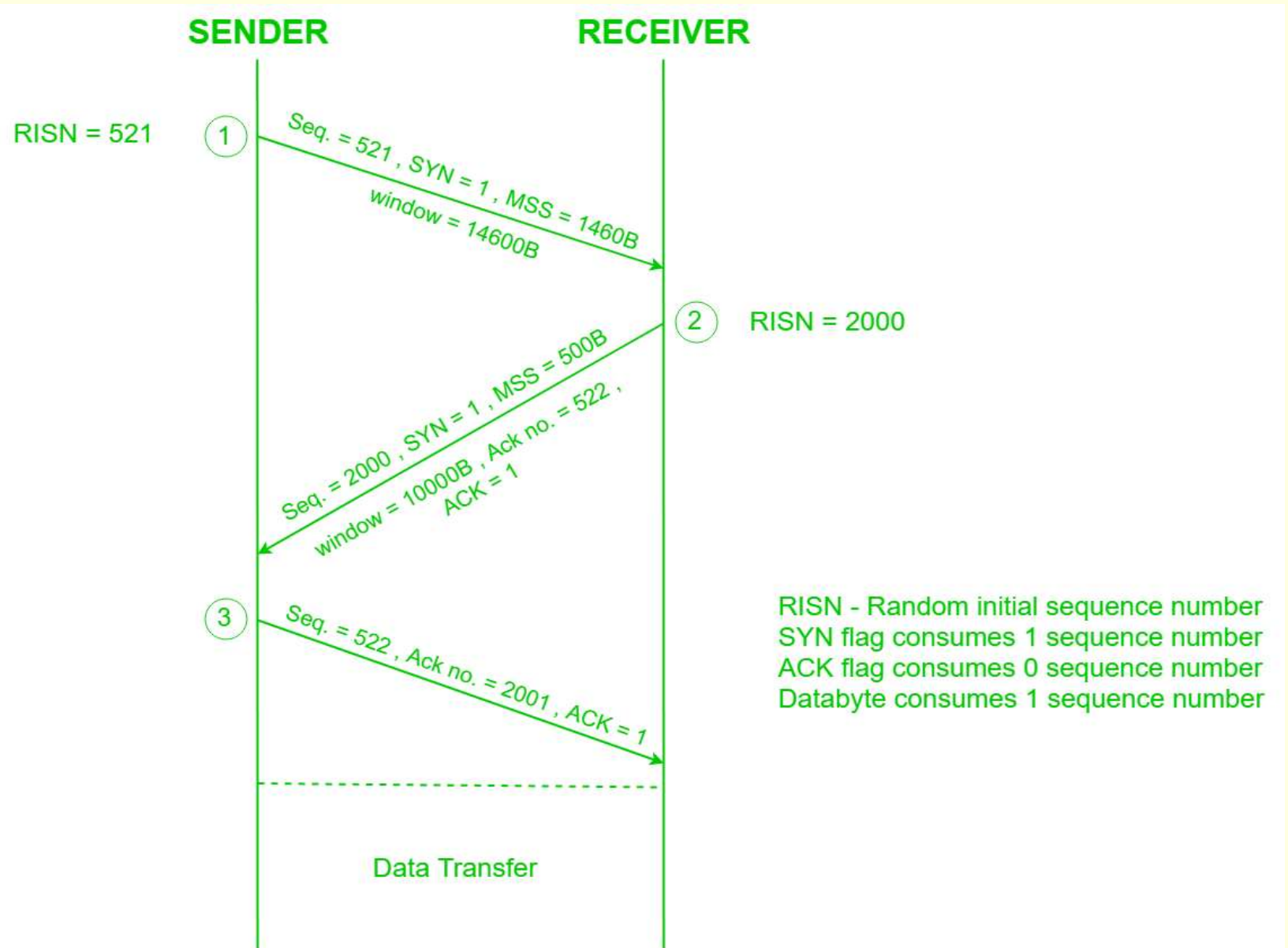
- An IP Transport Layer protocol is called Stream Control Transmission Protocol (SCTP).
- Many Internet applications use SCTP to perform transport layer duties, similar to User Datagram Protocol (UDP) and Transmission Control Protocol (TCP).
- On top of a connectionless packet network like IP, SCTP is a dependable transport protocol that facilitates data transfer over the network in scenarios involving one or more IP addresses.

Transport Layer Protocols:

■ Difference Between TCP and UDP at Transport Layer

TCP	UDP
TCP is a connection-oriented protocol	UDP is the connection-less protocol
TCP is reliable.	UDP is not reliable.
TCP supports error-checking mechanisms.	UDP has only the basic error-checking mechanism using checksums.
An acknowledgment segment is present.	No acknowledgment segment.
TCP is slower than UDP	UDP is faster, simpler, and more efficient than TCP.
Retransmission of lost packets is possible in TCP, but not in UDP.	There is no retransmission of lost packets in the User Datagram Protocol (UDP)
TCP has a (20-60) bytes variable length header.	The header length is fixed of 8 bytes.

TCP connection establishment:



TCP connection establishment:

- TCP is a connection-oriented protocol and every connection-oriented protocol needs to establish a connection in order to reserve resources at both the communicating ends.

- Connection Establishment –

1. Sender starts the process with the following:

- Sequence number (Seq=521): contains the random initial sequence number generated at the sender side.
- Syn flag (Syn=1): request the receiver to synchronize its sequence number with the above-provided sequence number.
- Maximum segment size (MSS=1460 B): sender tells its maximum segment size, so that receiver sends datagram which won't require any fragmentation. MSS field is present inside Option field in TCP header.
- Window size (window=14600 B): sender tells about his buffer capacity in which he has to store messages from the receiver.

TCP connection establishment:

2. TCP is a full-duplex protocol so both sender and receiver require a window for receiving messages from one another.

- **Sequence number (Seq=2000):** contains the random initial sequence number generated at the receiver side.
- **Syn flag (Syn=1):** request the sender to synchronize its sequence number with the above-provided sequence number.
- **Maximum segment size (MSS=500 B):** receiver tells its maximum segment size, so that sender sends datagram which won't require any fragmentation. MSS field is present inside Option field in TCP header.

Since $MSS_{receiver} < MSS_{sender}$, both parties agree for minimum MSS i.e., 500 B to avoid fragmentation of packets at both ends.

Therefore, receiver can send maximum of $14600/500 = 29$ packets. This is the receiver's sending window size.

- **Window size (window=10000 B):** receiver tells about his buffer capacity in which he has to store messages from the sender.

Therefore, sender can send a maximum of $10000/500 = 20$ packets. This is the sender's sending window size.

- **Acknowledgement Number (Ack no.=522):** Since sequence number 521 is received by the receiver so, it makes a request for the next sequence number with Ack no.=522 which is the next packet expected by the receiver since Syn flag consumes 1 sequence no.
- **ACK flag (ACK=1):** tells that the acknowledgement number field contains the next sequence expected by the receiver.

TCP connection establishment:

3. Sender makes the final reply for connection establishment in the following way:

- **Sequence number (Seq=522):** since sequence number = 521 in 1st step and SYN flag consumes one sequence number hence, the next sequence number will be 522.
- **Acknowledgement Number (Ack no.=2001):** since the sender is acknowledging SYN=1 packet from the receiver with sequence number 2000 so, the next sequence number expected is 2001.
- **ACK flag (ACK=1):** tells that the acknowledgement number field contains the next sequence expected by the sender.

Since the connection establishment phase of TCP makes use of 3 packets, it is also known as 3-way Handshaking (SYN, SYN + ACK, ACK).

TCP connection termination

- TCP supports two types of connection releases like most connection-oriented transport protocols:

- Graceful connection release

In the Graceful connection release, the connection is open until both parties have closed their sides of the connection.

- Abrupt connection release

In an Abrupt connection release, either one TCP entity is forced to close the connection or one user closes both directions of data transfer.

TCP connection termination

Abrupt connection release:

An abrupt connection release is carried out when an RST segment is sent. An RST segment can be sent for the below reasons:

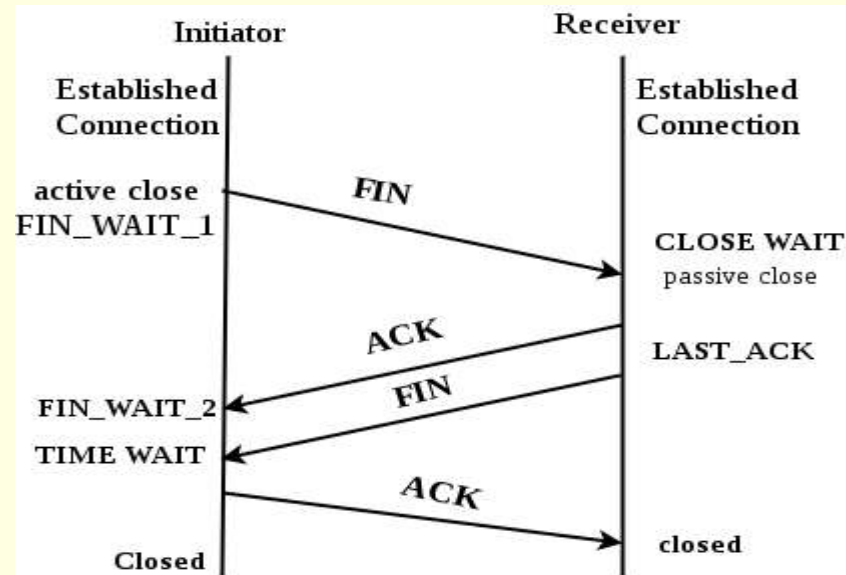
- When a non-SYN segment was received for a non-existing TCP connection.
- In an open connection, some TCP implementations send an RST segment when a segment with an invalid header is received. This will prevent attacks by closing the corresponding connection.
- When some implementations need to close an existing TCP connection, they send an RST segment. They will close an existing TCP connection for the following reasons:
 - Lack of resources to support the connection.
 - The remote host is now unreachable and has stopped responding.

When a TCP entity sends an RST segment, it should contain 00 if it does not belong to any existing connection else it should contain the current value of the sequence number for the connection and the acknowledgment number should be set to the next expected in- sequence number on this connection.

TCP connection termination

Graceful Connection Release:

- The common way of terminating a TCP connection is by using the TCP header's FIN flag. This mechanism allows each host to release its own side of the connection individually.



Step 1 (FIN From Client)

- Suppose that the client application decides it wants to close the connection. (Note that the server could also choose to close the connection). This causes the client to send a TCP segment with the FIN bit set to 1 to the server and to enter the FIN_WAIT_1 state. While in the FIN_WAIT_1 state, the client waits for a TCP segment from the server with an acknowledgment (ACK).

TCP connection termination

Step 2 (ACK From Server)

- When the Server received the FIN bit segment from Sender (Client), Server Immediately sends acknowledgement (ACK) segment to the Sender (Client).

Step 3 (Client waiting)

- While in the FIN_WAIT_1 state, the client waits for a TCP segment from the server with an acknowledgment. When it receives this segment, the client enters the FIN_WAIT_2 state. While in the FIN_WAIT_2 state, the client waits for another segment from the server with the FIN bit set to 1.

Step 4 (FIN from Server)

- The server sends the FIN bit segment to the Sender(Client) after some time when the Server sends the ACK segment (because of some closing process in the Server).

Step 5 (ACK from Client)

- When the Client receives the FIN bit segment from the Server, the client acknowledges the server's segment and enters the TIME_WAIT state. The TIME_WAIT state lets the client resend the final acknowledgment in case the ACK is lost. The time spent by clients in the TIME_WAIT state depends on their implementation, but their typical values are 30 seconds, 1 minute, and 2 minutes. After the wait, the connection formally closes and all resources on the client-side (including port numbers and buffer data) are released.

TCP Congestion Control

- **TCP congestion control** is a method used by the TCP protocol to manage data flow over a network and prevent congestion. **TCP uses a congestion window and congestion policy that avoids congestion.** Previously, we assumed that only the receiver could dictate the sender's window size. We ignored another entity here, the network. **If the network cannot deliver the data as fast as it is created by the sender, it must tell the sender to slow down.** In other words, in addition to the receiver, the network is a second entity that determines the size of the sender's window.
- **Congestion Policy in TCP**
 - **Slow Start Phase:** Starts slow increment is exponential to the threshold.
 - **Congestion Avoidance Phase:** After reaching the threshold increment is by 1.
 - **Congestion Detection Phase:** The sender goes back to the Slow start phase or the Congestion avoidance phase.

TCP Congestion Control

Slow Start Phase

- **Exponential Increment:** In this phase after every RTT the congestion window size increments exponentially.
- **Example:** If the initial congestion window size is 1 segment, and the first segment is successfully acknowledged, the congestion window size becomes 2 segments. If the next transmission is also acknowledged, the congestion window size doubles to 4 segments. This exponential growth continues as long as all segments are successfully acknowledged.

Initially $\text{cwnd} = 1$

After 1 RTT, $\text{cwnd} = 2^1 = 2$

2 RTT, $\text{cwnd} = 2^2 = 4$

3 RTT, $\text{cwnd} = 2^3 = 8$

TCP Congestion Control

Congestion Avoidance Phase

- **Additive Increment:** This phase starts after the threshold value also denoted as `ssthresh`. The size of CWND (Congestion Window) increases additive. After each RTT $\text{cwnd} = \text{cwnd} + 1$.
- **For example:** if the congestion window size is 20 segments and all 20 segments are successfully acknowledged within an RTT, the congestion window size would be increased to 21 segments in the next RTT. If all 21 segments are again successfully acknowledged, the congestion window size will be increased to 22 segments, and so on.

Initially $\text{cwnd} = i$

After 1 RTT, $\text{cwnd} = i+1$

2 RTT, $\text{cwnd} = i+2$

3 RTT, $\text{cwnd} = i+3$

TCP Congestion Control

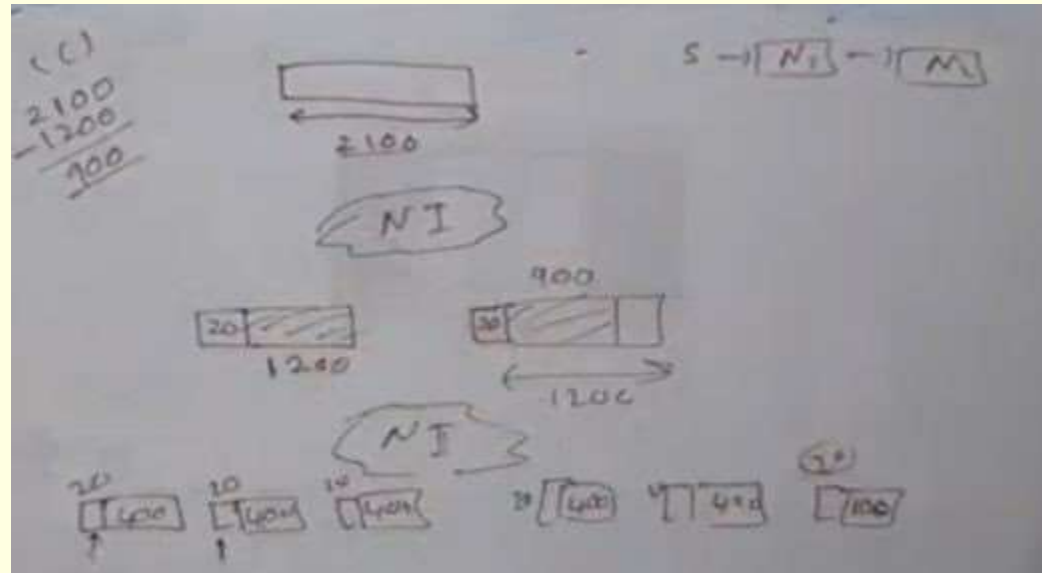
Congestion Detection Phase

- **Multiplicative Decrement:** If congestion occurs, the congestion window size is decreased. The only way a sender can guess that congestion has happened is the need to retransmit a segment. Retransmission is needed to recover a missing packet that is assumed to have been dropped by a router due to congestion. Retransmission can occur in one of two cases: when the **RTO timer** times out or when three duplicate ACKs are received.
- **Case 1:** Retransmission due to Timeout – In this case, the congestion possibility is high.
 - (a) ssthresh is reduced to half of the current window size.
 - (b) set cwnd = 1
 - (c) start with the slow start phase again.
- **Case 2:** Retransmission due to 3 Acknowledgement Duplicates – The congestion possibility is less.
 - (a) ssthresh value reduces to half of the current window size.
 - (b) set cwnd = ssthresh
 - (c) start with congestion avoidance phase

Practice Problem on TCP layer - 1

- A TCP message consisting of 2100 bytes is passed to IP for delivery across two networks. The first network can carry a maximum payload of 1200 bytes per frame and the second network can carry a maximum payload of 400 bytes per frame, excluding network overhead. Assume that IP overhead per packet is 20 bytes. What is the total IP overhead in the second network for this transmission?

Solution: $20 + 20 + 20 + 20 + 20 + 20 = 120$.

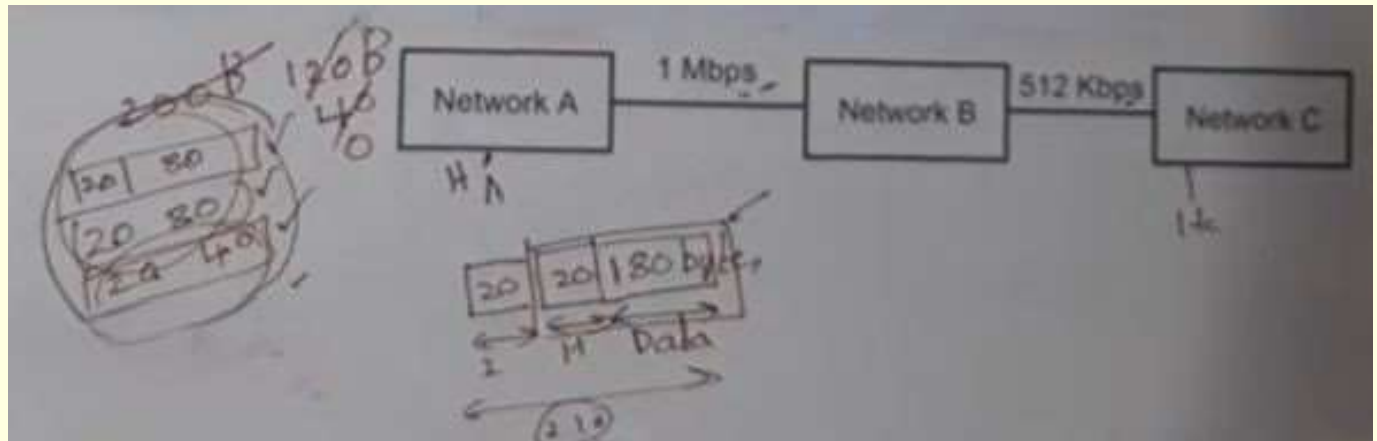


Practice Problem on TCP layer - 2

- Consider three IP networks A, B and C. Host HA in network A sends message each containing 180 bytes of application data to a Host HC in network C. The TCP Layer prefixes a 20 byte header to the message. This passes through an intermediate network B. The maximum packet size, including 20 byte IP header, in each network is A: 1000 bytes, B:100 bytes, C: 1000 bytes. The network A and B are connected through a 1 Mbps link, while B and C are connected by a 512 Kbps link. Assume that the packets are correctly delivered, how many bytes, including headers, are delivered to the IP layer at the destination for one application message, in the best case. (Consider only data packets).

Solution:

260



Application Layer:

Introduction:

- The **Application Layer** is topmost layer in the Open System Interconnection (OSI) model. This layer provides several ways for manipulating the data (information) which actually enables any type of user to access network with ease.
- This layer also makes a request to its bottom layer, which is presentation layer for receiving various types of information from it.
- The Application Layer interface directly interacts with application and provides common web application services.
- This layer is basically highest level of open system, which provides services directly for application process.

Functions of Application Layer:

Functions of Application Layer:

- Application Layer provides a facility by which users can forward several emails and it also provides a storage facility.
- This layer allows users to access, retrieve and manage files in a remote computer.
- It allows users to log on as a remote host.
- This layer provides access to global information about various services.
- This layer provides services which include: e-mail, transferring files, distributing results to the user, directory services, network resources and so on.
- It provides protocols that allow software to send and receive information and present meaningful data to users.
- It handles issues such as network transparency, resource allocation and so on.
- This layer serves as a window for users and application processes to access network services.
- Application Layer is basically not a function, but it performs application layer functions.

Functions of Application Layer:

- To ensure smooth communication, application layer protocols are implemented the same on source host and destination host.
- Features provided by Application Layer Protocols:
- The Application Layer protocol defines process for both parties which are involved in communication.
- These protocols define the type of message being sent or received from any side (either source host or destination host).
- These protocols also define basic syntax of the message being forwarded or retrieved.
- These protocols define the way to send a message and the expected response.
- These protocols also define interaction with the next level.

Application Layer Protocols:

Application Layer Protocols: The application layer provides several protocols which allow any software to easily send and receive information and present meaningful data to its users.

The following are some of the protocols which are provided by the application layer.

- **TELNET:** Telnet stands for Telecommunications Network. **This protocol is used for managing files over the Internet.** It allows the Telnet clients to access the resources of Telnet server. Telnet uses port number 23.
- **DNS:** DNS stands for Domain Name System. **The DNS service translates the domain name (selected by user) into the corresponding IP address.** For example- If you choose the domain name as www.abcd.com, then DNS must translate it as 192.36.20.8 (random IP address). DNS protocol uses the port number 53.

Application Layer Protocols:

Application Layer Protocols:

- **DHCP:** DHCP stands for Dynamic Host Configuration Protocol. **It provides IP addresses to hosts.** Whenever a host tries to register for an IP address with the DHCP server, DHCP server provides lots of information to the corresponding host. DHCP uses port numbers 67 and 68.
- **FTP:** FTP stands for File Transfer Protocol. **This protocol helps to transfer different files from one device to another.** FTP promotes sharing of files via remote computer devices with reliable, efficient data transfer. FTP uses port number 20 for data access and port number 21 for data control.
- **SMTP:** SMTP stands for Simple Mail Transfer Protocol. **It is used to transfer electronic mail from one user to another user.** SMTP is used by end users to send emails with ease. SMTP uses port numbers 25 and 587.

Application Layer Protocols:

Application Layer Protocols:

- **NFS:** NFS stands for Network File System. **This protocol allows remote hosts to mount files over a network and interact with those file systems as though they are mounted locally.** NFS uses the port number 2049.
- **SNMP:** SNMP stands for Simple Network Management Protocol. **This protocol gathers data by polling the devices from the network to the management station at fixed or random intervals,** requiring them to disclose certain information. SNMP uses port numbers 161 (TCP) and 162 (UDP).
- **HTTP:** HTTP stands for Hyper Text Transfer Protocol. It is the foundation of the World Wide Web (WWW). **HTTP works on the client server model. This protocol is used for transmitting hypermedia documents like HTML. This protocol was designed particularly for the communications between the web browsers and web servers,** but this protocol can also be used for several other purposes. **HTTP is a stateless protocol (network protocol in which a client sends requests to server and server responses back as per the given state), which means the server is not responsible for maintaining the previous client's requests. HTTP uses port number 80.**

Application Layer: Web and HTTP

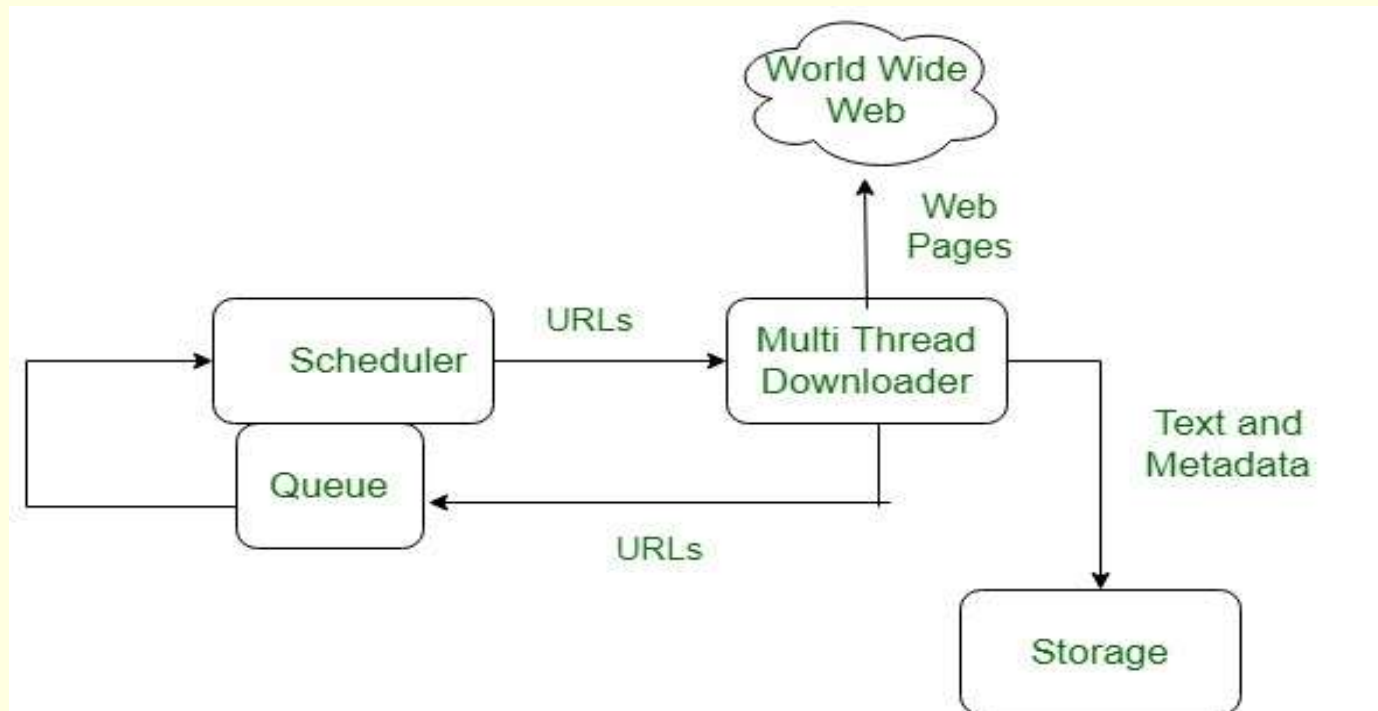
Web - Introduction:

- The World Wide Web (WWW), often called the Web, is a system of interconnected webpages and information that you can access using the Internet. **It was created to help people share and find information easily, using links that connect different pages together.** The Web allows us to browse websites, watch videos, shop online, and connect with others around the world through our computers and phones.
- From the user's point of view, the web consists of a vast, worldwide connection of documents or web pages. Each page may contain links to other pages anywhere in the world. The pages can be retrieved and viewed by using browsers of which internet explorer, Netscape Navigator, Google Chrome, etc. are the popular ones. The browser fetches the page requested interprets the text and formatting commands on it, and displays the page, properly formatted, on the screen.

Application Layer: Web and HTTP

Web – System Architecture:

- The browser is displaying a web page on the client machine. When the user clicks on a line of text that is linked to a page on the abd.com server, the browser follows the hyperlink by sending a message to the abd.com server asking it for the page.



Application Layer: Web and HTTP

Features of WWW:

- WWW is open source.
- It is a distributed system spread across various websites.
- It is a Hypertext Information System.
- It is Cross-Platform.
- Uses Web Browsers to provide a single interface for many services.
- Dynamic, Interactive and Evolving.

Components of the Web

There are 3 components of the web:

- **Uniform Resource Locator (URL):** URL serves as a system for resources on the web.
- **Hyper Text Transfer Protocol (HTTP):** HTTP specifies communication of browser and server.
- **Hyper Text Markup Language (HTML):** HTML defines the structure, organisation and content of a web page.

Application Layer: Web and HTTP

■ Difference between WWW and Internet:

WWW	Internet
It is originated in 1989.	It is originated in 1960.
WWW is an interconnected network of websites and documents that can be accessed via the Internet.	Internet is used to connect a computer with other computer .
WWW used protocols such as HTTP	Internet used protocols such as TCP/IP
It is based on software.	It is based on hardware.
It is a service contained inside an infrastructure.	There is a entire infrastructure in internet.

Application Layer: Web and HTTP

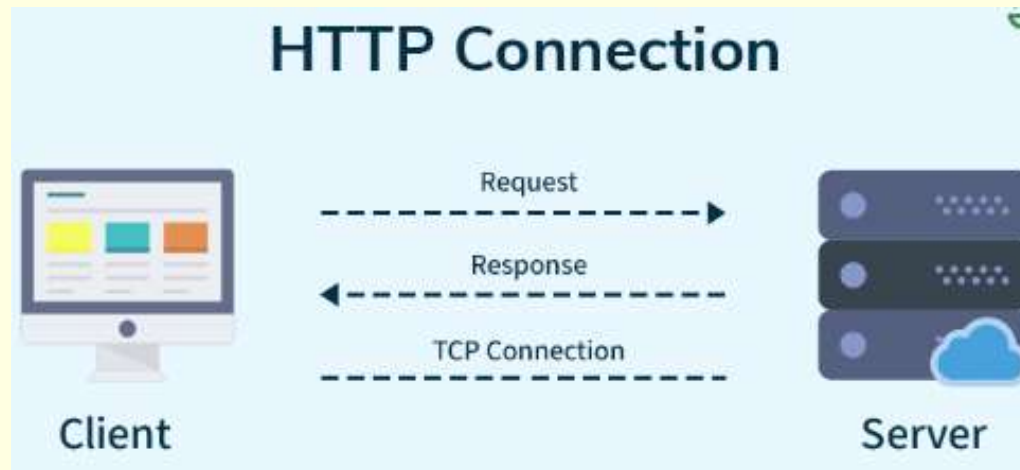
HTTP - Introduction:

- HTTP stands for HyperText Transfer Protocol. **It is the main way web browsers and servers communicate to share information on the internet.** Tim Berner invents it. HyperText is the type of text that is specially coded with the help of some standard coding language called HyperText Markup Language (HTML). HTTP/2 is the new version of HTTP. HTTP/3 is the latest version of HTTP, which is published in 2022.
- **When you visit a website, HTTP helps your browser request and receive the data needed to display the web pages you see.** It is a fundamental part of how the internet works, making it possible for us to browse and interact with websites.

Application Layer: Web and HTTP

Working of HTTP [HyperText Transfer Protocol]

- Whenever we want to open any website we first open a web browser after that we will type the URL of that website (e.g., `www.facebook.com`). This URL is now sent to the Domain Name Server (DNS). Then DNS first checks records for this URL in their database, and then DNS will return the IP address to the web browser corresponding to this URL. Now the browser is able to send requests to the actual server.



- After the server sends data to the client, the connection will be closed. If we want something else from the server we should have to re-establish the connection between the client and the server.

Application Layer: Web and HTTP

HTTP request is simply termed as the information or data that is needed by Internet browsers for loading a website. This is simply known as HTTP Request.

- There is some common information that is generally present in all HTTP requests. These are mentioned below:
 - HTTP Version
 - URL
 - HTTP Method
 - HTTP Request Headers
 - HTTP Body
- **HTTP Request Headers** generally store information in the form of key-value and must be present in each HTTP Request. The use of this Request Header is to provide core information about the client's information, etc.
- **HTTP Request Body** simply contains the information that has to be transferred. HTTP Request has the information or data to be sent to these browsers.
- **HTTP Methods** are simply HTTP Verb. In spite of being present so many HTTP Methods, the most common HTTP Methods are HTTP GET and HTTP POST. These two are generally used in HTTP cases. In HTTP GET, the information is received in the form of a website.

Application Layer: Web and HTTP

- **HTTP Response** is simply the answer to what a Server gets when the request is raised. There are various things contained in HTTP Response, some of them are listed below:
 - HTTP Status Code
 - HTTP Headers
 - HTTP Body
- **HTTP Response headers** are simply like an HTTP Request where it has that work to send some important files and data to the HTTP Response Body.
- **HTTP Responses body** are the responses that are received successfully upon the request. Generally, it comes under the requests generated by the web. In most cases, the request is to transfer the HTML data into a webpage.

Application Layer: Web and HTTP

- HTTP Status Codes are the 3-digit codes that tell the message or simply tell us about the HTTP Request whether it has been completed or not. There are simply 5 types of status codes:
 - Informational responses (100–199)
 - Successful responses (200–299)
 - Re-directional (300–399)
 - Client-Error (400–499)
 - Server-Error (500–599)

Practice Problem on HTTP server

- An IP machine Q has a path to another IP machine H via three IP routers R1, R2 and R3.

Q – R1 – R2 – R3 – H

H acts as an HTTP server, and Q connect to H via HTTP and download a file. Session layer encryption is used, with DES as the shared key encryption protocol. Consider the following four pieces of information:

[I1] The URL of the file downloaded by Q

[I2] The TCP port numbers at Q and H

[I3] The IP addresses of Q and H

[I4] The link layer addresses of Q and H

Which of I1, I2, I3, and I4 can an intruder learn through sniffing at R2 alone?

Solution:

Only I2 and I3

Practice Problem on HTTP server

- **Solution / Description:** Only I2 and I3

[I1] The URL of the file downloaded by Q

The intruder cannot see the URL because it is encrypted by the DES at the session layer.

[I2] The TCP port numbers at Q and H

The TCP port number **can be seen by the intruder** because we have TCP packet which include TCP header that contain both source and destination TCP port numbers.

[I3] The IP addresses of Q and H

The intruder can see because network layer, header contain IP addresses both source and destination .

[I4] The link layer addresses of Q and H

At Q, we are putting the address of R1 and Q & R1 will be having address of only Q. The intruder at R2 can see the address of R1; R3 only but not the address of Q; H (source and destination). Hence, intruder cannot see.

Application Layer: Email

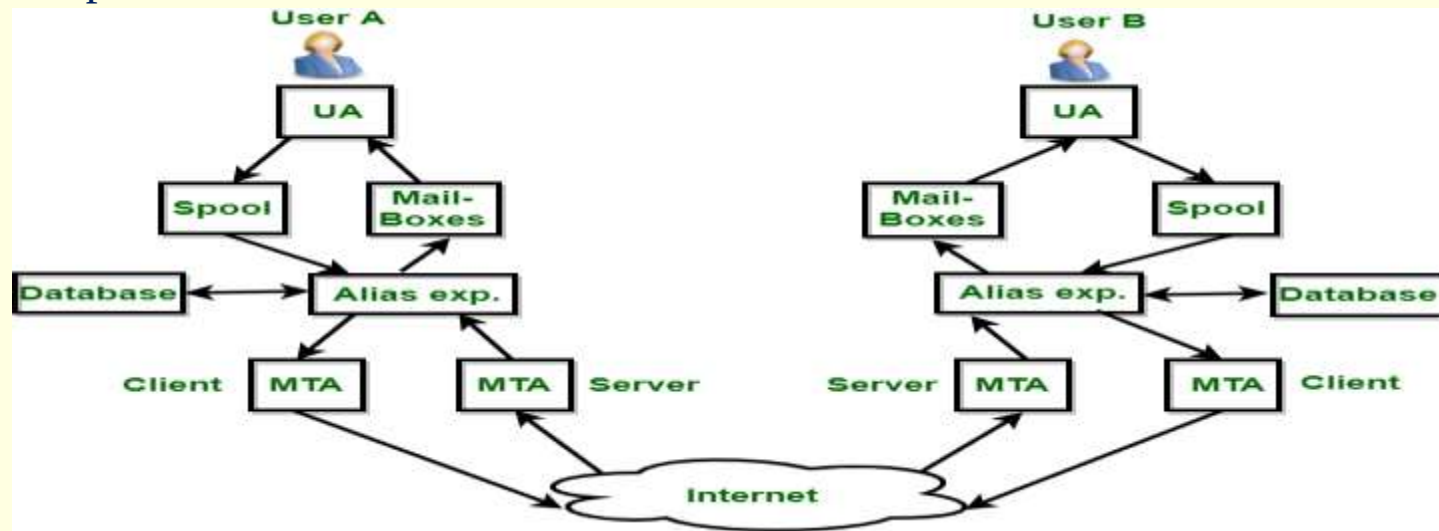
E-mail - Introduction:

- Electronic Mail (e-mail) is one of most widely used services of Internet. **This service allows an Internet user to send a message in formatted manner (mail) to the other Internet user in any part of world.** Message in mail not only contain text, but it also contains images, audio and videos data. The person who is sending mail is called sender and person who receives mail is called recipient. It is just like postal mail service.
- **Components of E-Mail System:** The basic components of an email system are: User Agent (UA), Message Transfer Agent (MTA), Mail Box, and Spool file.

Application Layer: Email

Components of E-Mail System:

- **User Agent (UA):** The UA is normally a program which is used to send and receive mail. Sometimes, it is called as mail reader. It accepts variety of commands for composing, receiving and replying to messages as well as for manipulation of the mailboxes.



- **Message Transfer Agent (MTA):** MTA is actually responsible for transfer of mail from one system to another. To send a mail, a system must have client MTA and system MTA. It transfer mail to mailboxes of recipients if they are connected in the same machine. It delivers mail to peer MTA if destination mailbox is in another machine. The delivery from one MTA to another MTA is done by Simple Mail Transfer Protocol.

Application Layer: Email

Components of E-Mail System:

- **Mailbox:** It is a file on local hard drive to collect mails. Delivered mails are present in this file. The user can read it delete it according to his/her requirement. To use e-mail system each user must have a mailbox. Access to mailbox is only to owner of mailbox.
- **Spool file:** This file contains mails that are to be sent. User agent appends outgoing mails in this file using SMTP. MTA extracts pending mail from spool file for their delivery. E-mail allows one name, an alias, to represent several different e-mail addresses. It is known as mailing list, Whenever user have to sent a message, system checks recipient's name against alias database. If mailing list is present for defined alias, separate messages, one for each entry in the list, must be prepared and handed to MTA. If for defined alias, there is no such mailing list is present, name itself becomes naming address and a single message is delivered to mail transfer entity.

Application Layer: Email

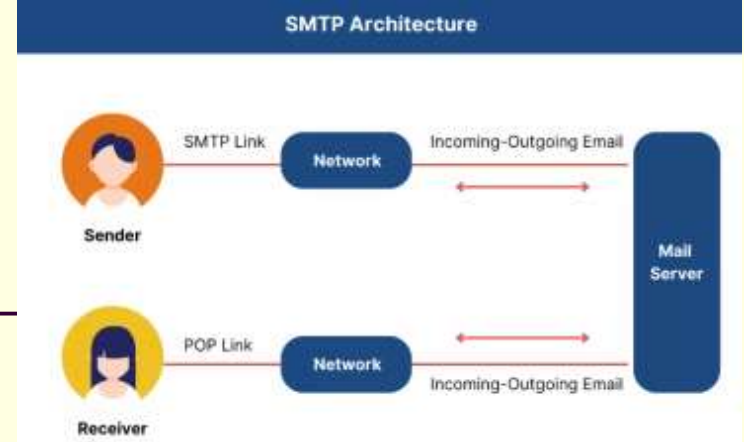
Services provided by E-mail system:

- **Composition** – The composition refers to the process that creates messages and answers. For composition any kind of text editor can be used.
- **Transfer** – Transfer means the sending procedure of mail i.e. from the sender to the recipient.
- **Reporting** – Reporting refers to confirmation for delivery of mail. It helps the user to check whether their mail is delivered, lost or rejected.
- **Displaying** – It refers to presenting mail in a form that is understandable by the user.
- **Disposition** – This step concerns with the recipient that what the recipient will do after receiving mail i.e., save mail, delete before reading or delete after reading.

SMTP

Simple Mail Transfer Protocol (SMTP):

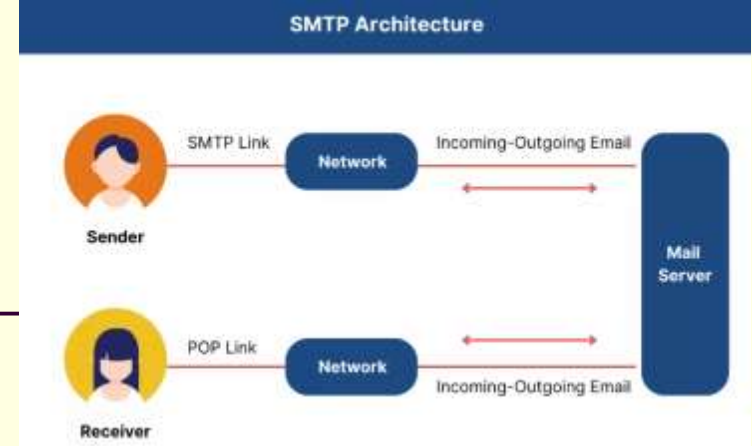
- SMTP (Simple Mail Transfer Protocol) is a standard protocol used in computer networks for sending and transferring email messages across the Internet. It operates at the application layer of the TCP/IP model and is responsible for transmitting emails from a sender's mail server to the recipient's mail server. SMTP typically uses TCP port 25 for communication, while port 587 is commonly used for secure email submission.
- SMTP follows a client-server model. The email sending process begins when a user composes a message using a User Agent (UA) such as an email client or webmail service. When the user sends the message, the email client connects to an SMTP server and transfers the message using the SMTP protocol.



SMTP

Simple Mail Transfer Protocol (SMTP):

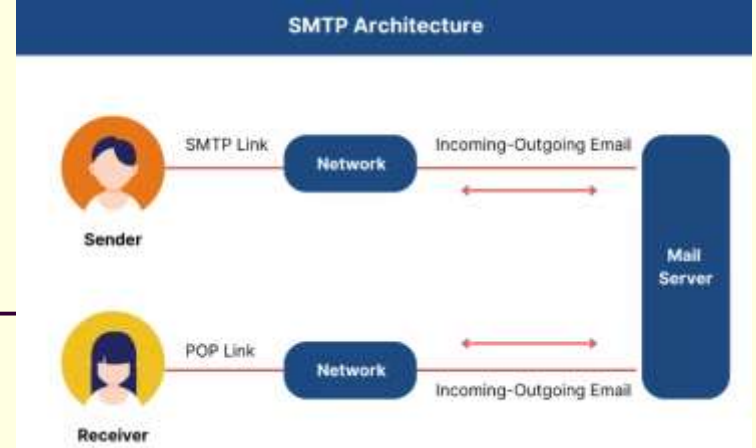
- The SMTP server then processes the recipient's email address to determine the destination mail server. It forwards the message through the network using a Mail Transfer Agent (MTA) until it reaches the recipient's mail server. Once the message arrives, it is stored in the recipient's mailbox by a Mail Delivery Agent (MDA).
- SMTP uses a set of text-based commands to communicate between servers. Common commands include HELO/EHLO to initiate communication, MAIL FROM to specify the sender's address, RCPT TO to specify the recipient's address, DATA to transmit the email content, and QUIT to terminate the session.



SMTP

Simple Mail Transfer Protocol (SMTP):

- It is important to note that SMTP is used only for sending emails, not for retrieving them. Email retrieval is handled by other protocols such as POP3 (Post Office Protocol) and IMAP (Internet Message Access Protocol).
- In summary, SMTP plays a crucial role in email communication by enabling reliable transmission of messages between mail servers across computer networks.



Application Layer: P2P network

- A peer-to-peer network is a simple network of computers. It first came into existence in the late 1970s.
- **Here each computer acts as a node for file sharing within the formed network. Also, each node acts as a server and thus there is no central server in the network. This allows the sharing of a huge amount of data.**
- The tasks are equally divided amongst the nodes. Each node connected in the network shares an equal workload. For the network to stop working, all the nodes need to individually stop working. This is because each node works independently.

Application Layer: P2P network

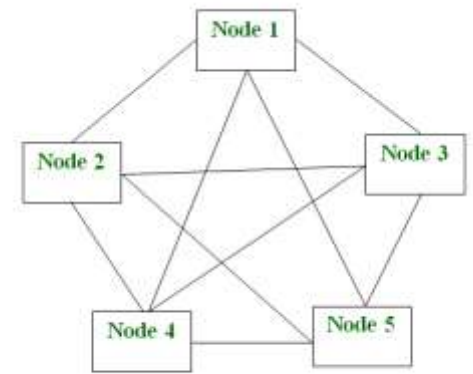
Types of P2P networks:

- **Unstructured P2P networks:** In this type of P2P network, each device is able to make an equal contribution. This network is easy to build as devices can be connected randomly in the network. But being unstructured, it becomes difficult to find content. For example, Napster, Gnutella, etc.
- **Structured P2P networks:** It is designed using software that creates a virtual layer in order to put the nodes in a specific structure. These are not easy to set up but can give easy access to users to the content. For example, P-Grid, Kademlia, etc.
- **Hybrid P2P networks:** It combines the features of both P2P networks and client-server architecture. An example of such a network is to find a node using the central server.

Features of P2P network:

- These networks do not involve a large number of nodes, usually less than 12. All the computers in the network store their own data but this data is accessible by the group.
- Unlike client-server networks, P2P uses resources and also provides them. This results in additional resources if the number of nodes increases. It requires specialized software. It allows resource sharing among the network.
- Since the nodes act as clients and servers, there is a constant threat of attack.
- Almost all OS today support P2P networks.

Application Layer: P2P network



P2P Architecture

■ P2P Network Architecture:

In the P2P network architecture, the computers connect with each other in a workgroup to share files, and access to internet and printers.

- Each computer in the network has the same set of responsibilities and capabilities.
- Each device in the network serves as both a client and server.
- The architecture is useful in residential areas, small offices, or small companies where each computer act as an independent workstation and stores the data on its hard drive.
- Each computer in the network has the ability to share data with other computers in the network.
- The architecture is usually composed of workgroups of 12 or more computers.

Application Layer: P2P Applications

- Applications of P2P Network:
- File sharing: P2P network is the most convenient, cost-efficient method for file sharing for businesses. Using this type of network there is no need for intermediate servers to transfer the file.
- Blockchain: The P2P architecture is based on the concept of decentralization. When a peer-to-peer network is enabled on the blockchain it helps in the maintenance of a complete replica of the records ensuring the accuracy of the data at the same time. At the same time, peer-to-peer networks ensure security also.
- Direct messaging: P2P network provides a secure, quick, and efficient way to communicate. This is possible due to the use of encryption at both the peers and access to easy messaging tools.
- Collaboration: The easy file sharing also helps to build collaboration among other peers in the network.

Application Layer: P2P Applications

Applications of P2P Network:

- **File sharing networks:** Many P2P file sharing networks like G2, and eDonkey have popularized peer-to-peer technologies.
- **Content distribution:** In a P2P network, unlike the client-server system so the clients can both provide and use resources. Thus, the content serving capacity of the P2P networks can actually increase as more users begin to access the content.
- **IP Telephony:** Skype is one good example of a P2P application in VoIP.

Computer Networks and Data Communication

